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Direct grafting of microwrinkled hydrogels by atmospheric-pressure plasma polymerization: Going simple and environmentally friendly

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ABSTRACT

The simple, direct fabrication of pH-responsive hydrogel coatings with microstructured wrinkles from a monomer mixture containing 2-hydroxyethyl methacrylate and 2-(diethylamino)ethyl methacrylate, grafted on silicon substrates via atmospheric-pressure plasma polymerization is reported. The process allows the synthesis of hydrogels with high conversion degree, which preserve the functionalities of both monomers as proven by XPS and FT IR analysis. Atomic force microscopy was conducted to characterize the wrinkle dimension, which revealed an increase of the wrinkle wavelength λ and amplitude A with the film's thickness. The chemical and morphological properties of the hydrogel films are preserved, even after immersion in deionized water for several days, indicating the high stability of the plasma-polymerized hydrogels. The simplicity of atmospheric-pressure plasma polymerization to deposit responsive hydrogels which are highly stable and having tunable properties as "materials on demand", will certainly extend the applications of plasma-based hydrogels especially for biomedical applications.

1. Introduction

Hydrogels are three-dimensional cross-linked polymer networks that have received remarkable attention in the last few decades because of their physical properties, the ability to resemble living tissues and their good biocompatibility [1,2]. Smart hydrogels are developed to respond to certain stimuli of physical, chemical, mechanical, biological or electrical nature, affecting their structures [1,2]. A Well exploited stimuli for hydrogels that are containing ionisable groups (e.g. COO– or R-NH_2) is the change of pH – acidic or alkaline – in a liquid environment [3]. For instance, a hydrophilic hydrogel network that provides a pH-sensitive swelling behaviour can be used for the transport of therapeutics for delivering and releasing [4] or for biosensors and tissue engineering [5–7]. Furthermore, a special feature of hydrogel coatings is the formation of folds or wrinkles on the surface, caused by surface instabilities during the fabrication process, in order to adopt an energetically more favorable state [8,9]. Wrinkled coatings offer significant advantages for sensor systems [10,11]. For example, the increased surface area allows an improved sensitivity and responsiveness to various stimuli, by accommodating a larger load of biomolecules and an increase of the signal-to-noise ratio [12,13]. Furthermore, a wrinkled microstructuration possesses inherent flexibility that enables to adapt and adhere to both rigid and flexible sensor substrates, ensuring proper contact and optimized sensor performance [14]. Guvendiren et al. [15] created swelling-induced wrinkle patterns in gradiently cross-linked HEMA hydrogels by simply varying the cross-linker volume concentrations and initial film thickness. In such a way, they have created a soft skin layer that gradually stiffened with depth. Zhao et. al. [16] fabricated the swelling-induced pattern in HEMA hydrogels for high-throughput generation of multicellular spheroids (MCS). The massive production of MCS they demonstrated may be used in cell biology and drug screening to help avoid the use of animals in research. Wrinkles were also explored to create densely arrayed nanofluidic channels [17], for smart adhesion [18] and were created on polymeric solar cells for a photocurrent enhancement up to 600% in the low-energy region of a model solar cell [19]. Recently, polymeric wrinkles were exploited for identity tags that mimics fingerprint [20].

Conventionally, hydrogels are fabricated via wet chemistry where monomer precursors, crosslinker agent, photoinitiator, and UV lamps are utilized. However, one of the major challenges is their poor adhesion to surfaces, which can limit applications. The contact between the surface and the analyte determines the performance of the sensor. Attempts

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were made to counteract this by modifying the surface properties and using adhesion promoters [21–23]. If, furthermore, a microstructured surface is needed for certain applications, an additional treatment step is necessary [24].

In the last decade, there is a growing interest on the preparation of stimuli-responsive materials using atmospheric-pressure plasma polymerization [25,26]. Plasma polymerization resembles free-radical photo polymerization involving free radicals and organic monomers in the production of polymers. The highly energetic and ionized environment of the plasma ensures that fast moving species initiate the cleavage of double bonds from the monomers and cross-link them together, without additional agents. Hence, the possibility of copolymerizing monomers of different chemical structures enables a broad range of newly, synthesized polymers. Atmospheric-pressure plasma polymerization is a technique that allows a simple deposition of monomers in very minuscule volume directly on substrates at ambient conditions. The avoidance of a multistep-procedure such as pre-cleaning substrates with piranha solution, surface-preinitiation with self-assembled monolayers and its subsequent initiator attachment and the utilization of protective environment conditions during the polymerization is another advantage. Furthermore, plasma polymerization can be performed on arbitrary substrate geometry, either flat or curved, with different surface chemistry, such as silicon or gold which are model substrates for surface immobilization, or other polymeric materials such as PET, PTFE, PEEK or PDMS, or even onto fabrics and nanopapers. Plasma treated sample surfaces are known to demonstrate improved adhesion properties through the incorporation of oxygen-containing functional groups [27,28]. In previous reports low-pressure or atmospheric-pressure plasma polymerization was utilized to generate various polymers immobilized on surfaces [29–31]. Bhattacharyya et al. [32] synthesized a new class of thin hydrogels from 1-amino-2-propanol (1A2P) and 2-(ethylamino)ethanol (2EAE) monomers via low-pressure pulsed plasma polymerization. The surface-attached hydrogels exhibit moisture uptake and thermoresponsive behavior. Especially, the 1A2P hydrogel changes color from blue at 20 °C to red at 80 °C. Tamirisa et al. [33] demonstrated a single step fabrication of thermoresponsive N-isopropylacrylamide (NIPAAM) hydrogels with tunable wettability below and above its lower critical solution temperature. A report about the fabrication of 2-(dimethylamino)ethyl methacrylate (DMAEMA) and 2-hydroxyethyl methacrylate (HEMA) hydrogels was made by Veuillet et al. [34]. By varying the power input, the mechanical properties of the surface-attached hydrogels can be tuned, 1 MPa to 4.5 MPa for HEMA and 4 MPa to 6 MPa for DMAEMA, respectively. However, the morphology and stability of the coatings after immersion in water as well as their responsive behavior, which is important for hydrogels, were not investigated. Recently, Alavi et. al. [35] showed, for example, a treatment of polymers made of polyethylene and poly-ε-caprolactone using an atmospheric pressure plasma jet to generate functional groups for biomolecule binding.

We have also previously demonstrated that the fabrication of hydrogels with the monomer combination of HEMA and DEAEMA by applying an atmospheric pressure plasma jet is possible [36]. For the first time, HEMA-co-DEAEMA hydrogels were generated. However, the morphology and stability of the coatings after immersion in water as well as their responsive behaviour were not investigated. To the best of the authors knowledge, this is the first study on the direct fabrication of micro-wrinkled hydrogel films by using atmospheric-pressure plasma. HEMA was selected as biocompatible monomer to improve the mechanical properties as well as transport properties whereas DEAEMA was chosen according to its pH responsive and swelling behavior. HEMA-containing hydrogels exhibit an adjustable elastic response and water content as tissue and shows excellent cytocompatibility. Determining the chemical structure and the topographical surface properties of the deposited hydrogel films depending on the mixture ratio was the major subject of the study.

2. Experimental

2.1. Materials

2-Hydroxyethyl methacrylate (HEMA, $C_6H_{10}O_3$, contains ≤ 250 ppm monomethyl ether hydroquinone as inhibitor, 97%) and 2-(Diethylamino)ethyl methacrylate (DEAEMA, $C_{10}H_{19}NO_2$, contains 1500 ppm MEHQ as inhibitor, 99%) were purchased from Sigma Aldrich, Germany and used as received. One-sided polished Silicon (Si) wafer with native oxide layer was bought from Microchemicals, Germany. Si wafer were cut into 1 cm × 1 cm substrates for hydrogel deposition. NIST-buffer solutions 4.01, 7.01 and 10.01 were used from Hanna Instruments GmbH, Germany.

2.2. Synthesis of HEMA-co-DEAEMA hydrogel coatings by plasma polymerization

The fabrication of the hydrogels using atmospheric pressure plasma polymerization is schematically illustrated in Fig. 1. For the synthesis, a radio frequency (27.12°MHz)-driven atmospheric-pressure plasma jet (detailed scheme and photograph in S1) operated at a plasma input power of 5 W and 1 standard litre per minute Argon (Ar) as working gas was employed [37]. HEMA:DEAEMA (HD) with different mixture ratios (vol./vol.) of 1:4 (HD14), 2:3 (HD23), 1:1 (HD11), 3:2 (HD32) and 4:1 (HD41) were used as monomer solution. Si substrate was first pretreated with Ar plasma for 30 s. Afterwards 0.1 μL of the liquid monomer mixture was dropped with a pipette on the substrate and Ar plasma was subsequently applied for 30 s to generate the hydrogel coating. The plasma process was performed at ambient conditions.

2.3. Characterization

IR spectra was recorded using Bruker Vertex 70 V (Germany) to get the information on the functional groups of the hydrogels. The measurements were performed in transparent mode for 100 scans per measurement with resolution of 4 cm^{-1}. The transmittance spectra were recorded.

Surface elemental composition was investigated by means of an X-ray photoelectron spectrometer (XPS Axis Supra DLD electron spectrometer, Kratos Analytical, Manchester, UK) with ESCApe software acquisition. The X-ray source was a monochromatic Al Kα source (1486.6 eV), operating at 150 W with an analysis area of approximately 250 μm in diameter. Data processing was carried out using CasaXPS software, version 2.14dev29 (Casa Software Ltd., Teignmouth, UK) wherein an adjustment of peaks with a Gaussian-Lorentzian shape and Shirley background subtraction was performed.

The zeta potential was measured by means of an electrokinetic analyser (SurPASS 3, Anton Paar GmbH, Ostfildern, Germany) equipped with an adjustable gap cell (2 cm in length, 1 cm in width and ~ 100 mm in height). The surface zeta potential was determined in function of pH in a 1 mM KCl electrolyte solution varying the solution pH from 4 to 9 by adding 0.05 M HCl or 0.05 M NaOH through the instruments automatic titration unit. Separate couples of samples were used for the acidic and basic titrations in order to avoid artifacts due to surface reactions during the measurement. Three measurements were carried out for each pH point. The zeta potential was calculated according to the equations of Helmholtz-Smoluchowski [38].

Atomic force microscopy (AFM) (coreAFM, Nanosurf, Switzerland) was used to obtain the surface morphology of the hydrogels, of which the characteristic of the wrinkles (wavelength, λ, and amplitude, A) can be determined, as well as the root-mean-square surface roughness (rms). AFM images were acquired in non-contact mode, using PPP-XYNCHR tip with spring constant of 42 N/m. AFM images were processed using the Gwyddion freeware.

A surface profiler (Dektak3St Surface profiler from Bruker, Germany) was used to measure the thickness of the hydrogel coatings.

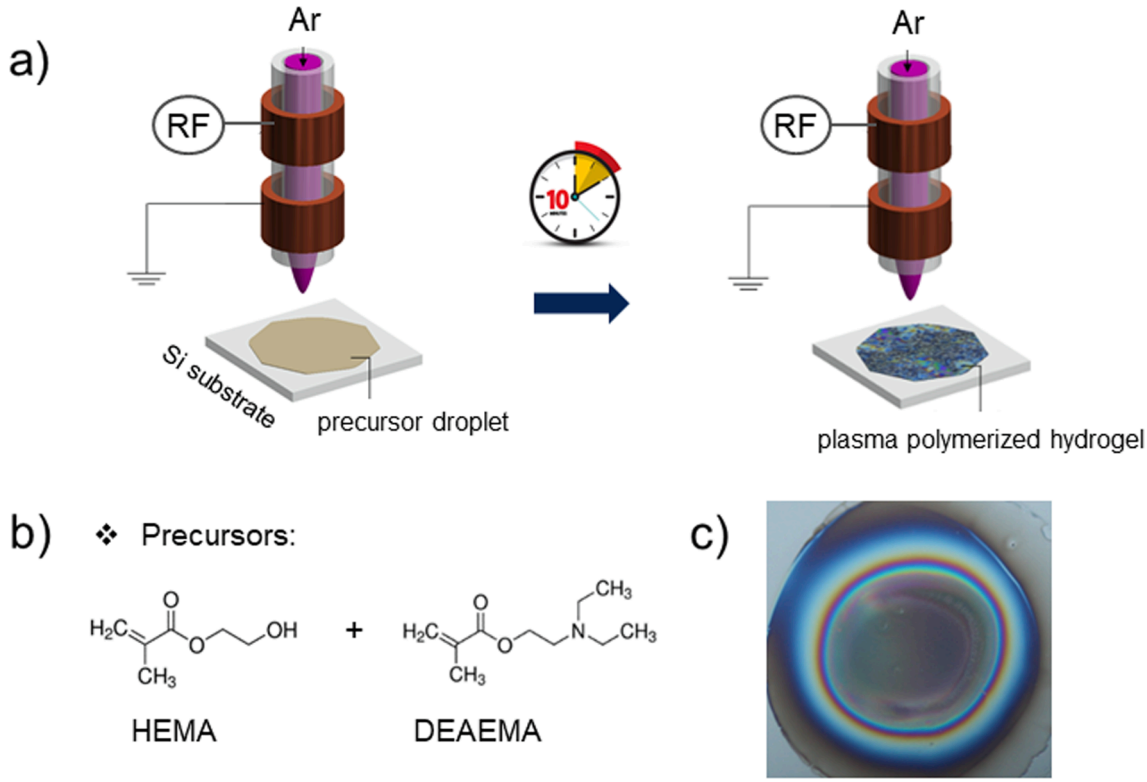

Fig. 1. a) A schematic illustration of the surface-immobilized plasma hydrogel via atmospheric pressure plasma polymerization, b) the chemical structures of HEMA and DEAEMA. c) a photograph of a surface immobilized HD hydrogel on Si wafer.

3. Results and discussion

3.1. Synthesis and chemical characterization

Plasma-induced polymerization is a solvent-free technique enabling a deposition of polymers or hydrogels directly onto the surfaces of the substrates, without additional cross-linking agents. According to the plasma polymerization process, radicals initiate the layer formation process. Furthermore, through the interaction of electrons, ions and electromagnetic radiation fragmentation, recombination and/ or rearrangement of one or more monomers take place [39,40]. The result is a disordered, highly cross-linked plasma polymer with various functional groups and significant differences in contrast to conventional polymers [39,40].

For the deposition of hydrogel films, Si wafer needs a pre-treatment in a first step to improve the surface wettability, whereby the monomer mixture droplet spreads on the surface due to the hydrophilic nature of the activated substrate (figure S2). The monomer mixtures used were HEMA(H): DEAEMA(D) 4:1 (vol/vol) (HD41), HD32, HD11,HD23 and HD14, respectively. The divergence of the droplets is due to the different viscosities of HEMA (6.8 mPa·s at RT) and DEAEMA (1.5 mPa·s at RT). The HD41 monomer mixture thus shows the lowest flowability. Atmospheric pressure plasma was then applied to the droplet, starting the polymerization process. To find the best deposition conditions, process parameters like gas flow, input power, working distance and treatment time were varied systematically beforehand. A milky-turbid layer with characteristic, colored interferences [41,42] indicated a hydrogel-like layer (Figure S3). All coatings with this appearance were optically tested with regard to delamination and solubility in aqueous medium. Hydrogel coatings with shorter fabrication times were either unstable after washing with aqueous media or did not have a good crosslinking degree (partial delamination or cracks within the coating). The other case are coatings that are generated with longer polymerization times (longer plasma application). A general characteristic of (any) hydrogel is that the speed of the reaction depends on the size, shape, degree of crosslinking, and the number of ionic groups in the gel [43]. The response rate increases the smaller, thinner, and less crosslinked the hydrogel is and the more ionic groups it inherits [43–45]. That means longer polymerization time leads to higher crosslinking, but also less functionality in terms of flexibility (swelling/shrinking), biomolecule interaction, and stimuli-responsive behavior [46]. In terms of suitability for a broad range of biomedical applications, the stability of the immobilized hydrogels is very important. Figure S4 shows the mass-left ratio (determined by weighing, as deposited and after immersion of the samples) of the hydrogels after immersion in water for 24 h and afterwards dried for 6 h. The total initial mass loss was ~ 30 %, depending on the mixture ratio. We observed no further mass loss after immersion in water for the next 5 days (figure S5). As bulk monomers are used directly during the fabrication, it is expected that amount of the unbound physisorption polymers is present in the hydrogels. The 24 h of immersion is an important step to remove them. Once the unbound polymers were washed away, only stable and covalently bound hydrogels remain.

The chemical structure of all HD hydrogels was analysed as-deposited (Fig. 2) and after immersion in deionized water for 24 h (figure S6) by FT-IR spectroscopy. HEMA and DEAEMA are vinyl monomers, which have hydroxyl and (tertiary) amine functional groups, respectively. According to the monomer's structure, FT-IR spectra revealed a broad absorption band in the region of 3000–3600 cm^{-1} with a maximum at 3450 cm^{-1}, originating from the O–H stretching vibration in hydroxyls of HEMA, the characteristic and strong absorption peak at 1730 cm^{-1} assigned to C = O stretching vibration of ester groups, while C-O stretching and C–H bending vibrations were found to be in the region between 1500 and 700 cm^{-1}, in particular C–H bending vibrations at 1457 cm^{-1} and C–O stretching vibration of ester groups at 1160 cm^{-1}, respectively. C = C stretching vibration were detected at 1638 cm^{-1}. The presence of aliphatic chains is demonstrated in the spectral range of 2970–2840 cm^{-1}. In particular, the absorption band at 2937 cm^{-1} is

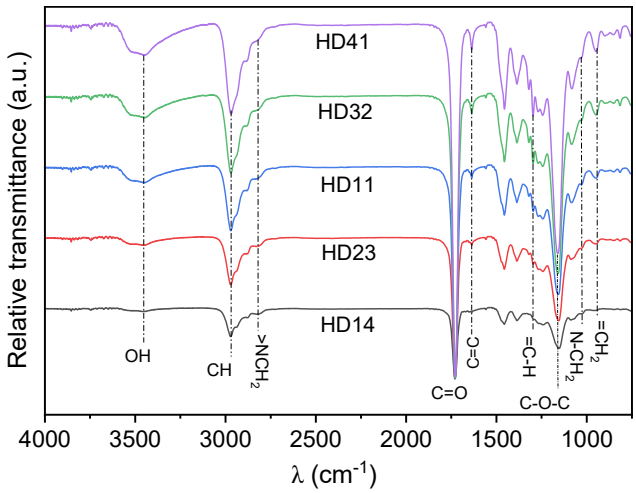

Fig. 2. Transmittance FT-IR spectra of plasma-polymerized HD films depending on the mixture ratio.

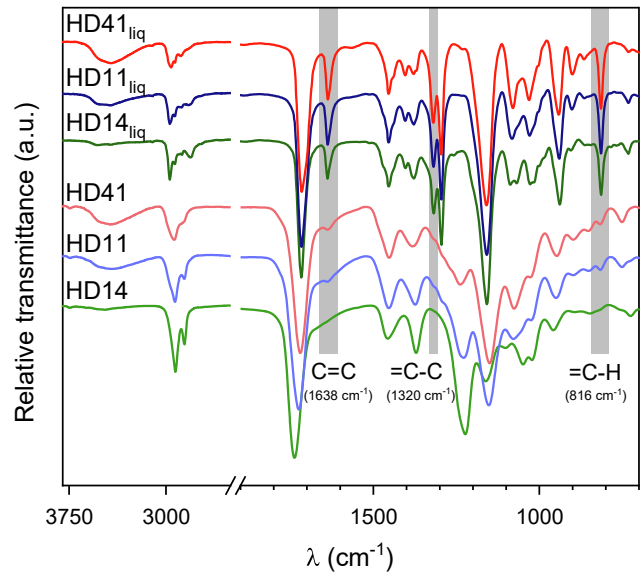

Fig. 3. Transmittance FT-IR spectra of non-polymerized $HD41_{liq}$, $HD11_{liq}$ and $HD14_{liq}$ as well as plasma polymerized HD41, HD11 and HD14 films, respectively.

resulting from the C–H stretching vibrations of CH_3 and CH_2 groups. The broad OH signal indicates the presence of inter/intramolecular hydrogen bonds, suggesting the existence of ordered domains in the copolymer [47]. The deformation vibrations of $–CH_2$- and $–CH_3$ groups occur in the spectra at 1472 cm^{-1}. Another significant band occurring at 1296 cm^{-1} can be assigned to out-of-plane C–H bending (methylene twisting) in aliphatic chains. The FT-IR data obtained revealed that HD hydrogels retained functionalities of both monomers: OH (3300–3600 cm^{-1}), a broadened > NCH_2 peak (~2800 cm^{-1}), C = O (1730 cm^{-1}), and a broadened C-O-C peak at 1160 cm^{-1}. Among the HD hydrogels studied, differences in the molecular structure in terms of the signal intensity of peaks at 1638 cm^{-1} and 1320 cm^{-1} assigned to C = C and = C–H, respectively, were observed. For HD41, the peaks were more intense as compared to HD14 where the peaks almost disappeared. This indicates that the plasma-based polymerization mechanism of HD mixtures is facilitated by the cleavage of vinyl groups. The average kinetic electron energy of the plasma is ~ 3 eV [39] and the dissociation energy of the π-bond of $H_2C = CH_2$ is 2.7 eV [48], therefore the probability of reactions via free radical species at this point is enhanced, leading to a decrease of related peaks. Furthermore, the activation energy needed to initiate chain growth polymerization of vinyl monomers is about 1 eV [39]. This observation is also in agreement with the mechanism reported in other works [34,49,50]. For a better comparison, Fig. 3 shows FT-IR spectra of non-polymerized HD mixtures and their hydrogel counterparts. A remarkable decrease in the signals assigned to C = C and = C-O was observed. Moreover, the C = O stretching band at 1718 cm^{-1} is shifted to 1730 cm^{-1} for the plasma polymerized HD hydrogels, indicating the lack of the conjugated ester groups (presence of α,β-unsaturated ester groups). The decrease in the intensity of the bands at 1637 cm^{-1} and 816 cm^{-1} indicates the change from vinylic to methylenic carbons. The peak at 750 cm^{-1} is due to CH_2 of methacrylic polymers [51]. These findings are evidences of the occurrence of the polymerization reactions through the C = C bonds, associated to the loss of conjugation for the C = O bond as well as cross-linking reactions [52]. Furthermore, compared to the non-polymerized HD mixtures, spectra of HD films showed a peak between 1200 and 1100 cm^{-1} attributed to C–O stretching.

No changes in the chemical structure as well as in the presence of functional groups were observed after immersion in water for 24 h and even after 7 days (figure S4). This result evidences the good stability of the hydrogel films in aqueous environment and indicates that both functionalities of HEMA and DEAEMA in HD hydrogels are preserved.

Since the plasma polymerization is mediated by the cleavage of the vinyl double bond, the degree of conversion of the hydrogel films can be calculated by comparing the vinyl peak (C = C) before and after the synthesis versus the internal standard (C = O) [53]. The conversion degree for as-deposited HD1:4, HD2:3, HD1:1, HD3:2 and HD4:1 are 94%, 94%, 93%, 93%, and 90%, respectively. After immersion in water, the values slightly increase to same level, 95%, indicating that the unbound polymers are washed away during the immersion.

To get further information on the elemental composition of the HD coatings, XPS was conducted before and after immersion in deionized water. As shown in Fig. 4, the oxygen to carbon ratio, O/C, is increasing with increasing HEMA portion whereas the nitrogen to carbon ratio, N/C, is decreasing. In other word, the higher the DEAEMA in the mixture, the higher the N/C ratio is. Upon immersion, the values are slightly lower due to the removal of non-polymerized monomer or unbound polymers. The XPS results clearly evidence the incorporation of HEMA and the pH-responsive DEAEMA in the hydrogels. The obtained results are in good agreement with those reported by Gonzalez-Henriquez [24]. A more detailed chemical investigation of the HD hydrogels has been carried out by analysing the high resolution C1s spectra (Table 1). C1s

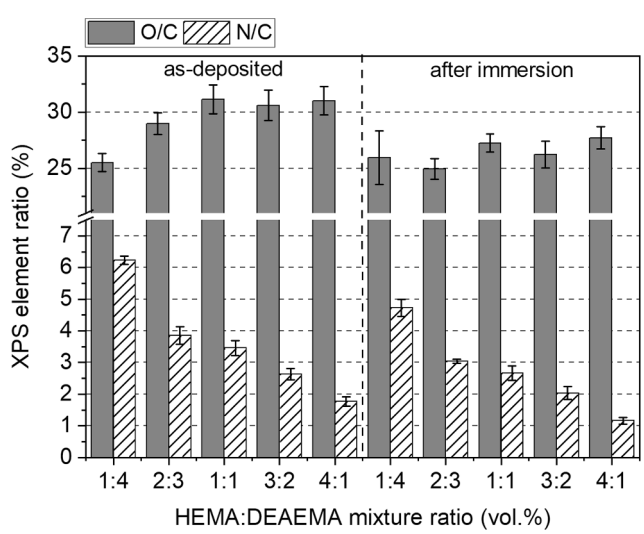

Fig. 4. Elemental composition of HD coatings as-deposited and after immersion in deionized water (n = 3).

Table 1
Attributions, binding energy values and atomic percentages of XPS C1s signals fitting components determined for HD coatings as-deposited (ad) after immersion in water (im) depending on the mixture ratio.

	C–C/H 285.0 eV	CC(=O)OR C-N 285.8 eV	C-O 286.5 eV	COC = O 287.1 eV	C(=O)O 289.2 eV
HD14ad	48.2	20.7	10.4	10.9	9.4
HD14im	56.6	14.6	12.3	7.1	9.2
HD23ad	56.6	9.5	13.5	9.2	10.9
HD23im	54.5	16.5	10.3	9.4	9.0
HD11ad	57.6	6.8	13.2	10.2	12.1
HD11im	58.3	9.4	14.0	8.0	10.1
HD32ad	58.0	7.0	13.5	9.4	11.9
HD32im	58.6	11.9	11.7	8.4	9.2
HD41ad	60.3	5.3	13.0	9.6	11.6
HD41im	60.6	7.9	12.7	8.7	9.9

signals were fitted by five major components at binding energies of 285.0, 285.8, 286.5, 287.1 and 289.2 eV, which can be attributed to aliphatic C–H/C–C, C-CO-O/C-N, C-O, C = O/O-C-O and O-C = O [54,55] (survey spectra and fitted C1s spectra can be found in figure S7 and S8). The full width at half maximum (FWHM) and the Gaussian/Lorentzian ratio were fixed at 1.2–1.4 eV and 30%. The most significant differences are the proportions of C-N with higher DEAEMA content in the hydrogel mixture. While the as-deposited value for HD14 is ~ 20%, the C-N portion decreases for all of the other mixtures, reaching a value of ~ 5% for HD41. However, after storage in water, the values for O/C ratio are similar for all mixtures that no remarkable trend can be observed. The C-O-C = O proportion decreases for all HD films after immersion, indicating again the removal of unbound monomer units. At least in the cases of HD14 and HD11 the C-O content increases up to 2 %, which can be explained by the introduction of terminal oxygen-containing groups into the plasma polymerized hydrogel films due to storage in H_2O [56]. Interestingly, for N/C ratios, after immersion the trend is the same to that of the as-deposited, where HD14 has the highest N/C ratio and HD41 is the lowest. Overall, no significant changes in the chemical structures were observed, between the as-deposited hydrogels and the after immersion hydrogels. In agreement with FT-IR, the XPS results thus confirm the good stability in an aqueous environment and the preserved functionalities of HD hydrogels.

3.2. Zeta potential

To get more information on the surface functionality and surface charge properties of the hydrogels, zeta potential (ζ) measurements were performed. The zeta potential describes the surface charging behavior in contact with water-based electrolytes (in function of the pH value) and gives further information on the isoelectric point. Thus, information of the functional groups exposed at the solid–liquid interface are provided. Furthermore, the ζ- potential directly influences the stability of suspensions and gives indications for the adhesion between solids. In addition, the adsorption and chemical reaction between solids and ions/ molecules can be observed. To relate the electrokinetic phenomena of the streaming potential to the zeta potential the Helmholtz-Smoluchowski approximation was used [38]. For this purpose, HD hydrogel films of different mixture ratios were deposited on 3 cm × 2.5 cm glass microscope slide. Zeta potential analysis was carried out from pH 4 to pH 9, with increment of one pH value. For each pH, three measurements were performed and the average value was calculated (Fig. 5). As reference surface, measurements on uncoated glass microscope slides (also three samples for each pH value, mean value was taken) were performed. Material surfaces without functional groups behave more hydrophobic, due to polarization effects of water molecules and/or ion replacement, which is described more in detail here [57–59]. Gu et al. [60] found zeta values for glass slides in the range from − 38.7 to − 84.2 mV as the pH value increases from 2.68 to 10.32.

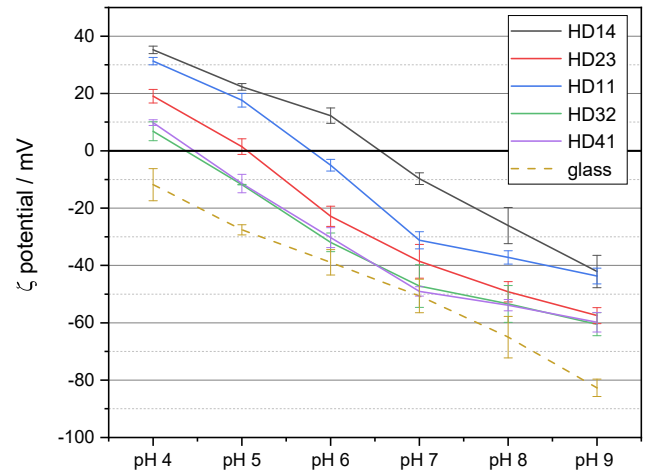

Fig. 5. Zeta potential measurements of the plasma-polymerized hydrogel mixtures in function of different pH values, with glass as reference (n = 3).

A lower pH value will increase the concentration of H^+ (from HCl) in the streaming solution, while a higher pH value causes OH^- (from NaOH) to be the potential-determining ions [60]. OH^- and H^+ ions are strongly adsorbed in the region near the glass–water interfaces. Interestingly, in this study even at pH 2.68 the ζ-potential is still negative, suggesting that the selective adsorption of H^+ is not strong enough to counteract the strongly negative ζ-potential [60]. We observed the same trend with similar values for glass as reference shown in Fig. 5.

In case of the HD coatings that contain functional groups, acid-basic reactions between the liquid environment and these groups occur, including for example protonation of the tertiary amine groups. In general, the tertiary amine groups of DEAEMA are protonated at a pH below its pK_a value of 7.3, whereas a pH above causes a deprotonation [61–63].

For HD14 layers this results in positive zeta values at lower pH, until the IEP (isoelectric point) - zeta potential equals to zero -, of 6.5 ± 0.4 is reached. For HD11 films, comparable zeta potentials can be observed, showing positive values for pH 4 and 5 with an IEP at 5.9 ± 0.1, whereby a higher pH leads to negative zeta values. Also for HD23 negative zeta values are reached as early as pH 6 and above, the IEP is lower with 5.0 ± 0.2. Surprisingly, the IEP for HD11 is higher with 5.8 ± 0.2, which might be a result of randomly distributed, recombined and rearranged fragments during the plasma-polymerization, leading to the fact that not all amine groups from HD23 were detected with the ζ-potential measurements. Nevertheless, the trend is the same. A high pH value will enhance the adsorption of hydroxyl groups onto the surface and at the same time suppress the protonation of amine groups. This effect is evident for the HD32 and HD41 film with the lowest amount of DEAEMA (lowest level of ionization), producing negative zeta values even at significantly lower pH values. Only at pH 4 positive zeta values of ~ 9 mV are generated with similar IEPs of 4.4 ± 0.2 and 4.5 ± 0.1, respectively. These results evidence that our hydrogels are endowed with functional groups, which can be protonated/deprotonated and thus be stimulated by solutions with different pH values.

3.3. AFM measurements

Microstructured wrinkles on the surface of electrodes or sensors can improve the biocompatibility, and increase the immobilization of biomolecules as well as the analytical sensitivity [64–66]. This is caused by the increased ratio of surface area to volume, which can lead to more binding capabilities and an improved accessibility to the analyte or substrate [67]. AFM analysis was performed to get information on the topographical properties of the hydrogel films. Here, the influence of the monomer mixture on the total thickness and wrinkle size, namely

wrinkle wavelength λ and amplitude/ height A, is investigated. AFM images, shown in Fig. 6, reveal the formation of a micro-wrinkled surface pattern. It can be clearly seen that the characteristics of the wrinkled micropattern are influenced by the HD mixture ratio. Wrinkle phenomena was first reported by Tanaka et al. [68] when they reported the formation of swelling-induced wrinkles on acrylamide-co-sodium acrylate hydrogels. They proposed that the initial gel fixed on the substrate is mechanically constrained while the top layers are free to expand, generating an anisotropic osmotic pressure along the film thickness. Due to the mismatch and competition between the mechanical constraints and osmotic pressure, wrinkles are formed. AFM micrographs of the HD hydrogels exhibit wrinkles of different size depending on the mixture ratio of the HD monomers. More precisely, an increase of the wrinkle size with increasing HEMA portion can be observed. This is in correlation with the filḿs thickness that increases along with the HEMA portion, equally with R_{RMS}, as shown in Table 2. By applying different HD mixture ratios, fully closed hydrogel films with thicknesses ranging from 400 to 1600 nm were synthesized. The wrinkle wavelength, λ, and amplitude, A, according to the different mixture ratios are depicted in Fig. 7. For as-deposited hydrogel films, by decreasing the amount of DEAEMA (increasing the HEMA portion) in the mixture ratio, λ and A are increasing. In particular, λ is increasing from 1.39 ± 0.11 μm for HD14 up to 9.46 ± 1.09 μm for HD41 while A increases from 0.12 ± 0.01 μm to 1.58 ± 0.14 μm, respectively. The R_{RMS} increases from 35 ± 1 nm for 1:4 to 539 ± 37 nm for 4:1. Interestingly, after immersion in water for 24 h, the wrinkle characteristics are comparable with a slight decrease of λ and A due to the loss of non-polymerized monomers or unbound polymers. Moreover, the ratio of the amplitude versus the wavelength of the wrinkles which may indicate the compactness of wrinkles, follows also a similar trend, increasing from HD14 to HD41 where the value for each mixture is slightly lower after immersion (Fig. 7). The obtained results indicate the direct influence of the mixture composition on the total thickness of the films, which consequently influences the wrinkle wavelength and amplitude. Furthermore, AFM analysis evidenced that the wrinkle characteristics, which reflect the physical properties of the hydrogel films, are preserved after immersion in water. The wrinkle characteristics appear to change with thickness just as much as with the different ratio of the monomers. For that reason HD14 and HD41 films with a comparable thickness were deposited by adjusting the volume of the monomer to further analyse if any changes in the wrinkle pattern is induced. As seen in figure S9, a thickness of 480 ± 15 nm was determined for the as-deposited HD14 and HD41 films. Furthermore, both HD films exhibit comparable wrinkle characteristics with $\lambda = 2.37 \pm 0.34$ μm and $A = 0.23 \pm 0.04$ for HD14 and $\lambda = 2.39 \pm 0.24$ μm and $A = 0.23 \pm 0.04$ μm for HD41, respectively. After immersion (figure S10), for HD14, the values are 1.97 ± 0.39 μm and 0.23 ± 0.01 μm, for λ and A, respectively. For HD41 2.46 ± 0.27 μm and 0.18 ± 0.03 μm were observed for λ and A.

While there is a clear correlation between the morphology of wrinkles and the HD mixtures, their responses to different condition and mechanical properties, are beyond the scope of this paper. Detail investigation is on progress and will be reported separately.

Table 2
Film thickness and root mean square roughness, R_{rms}, of as-deposited HD films.

coating	thickness (μm)	R_{rms} (nm)
HD14	0.470 ± 0.039	34.92 ± 1.17
HD23	0.647 ± 0.077	144.13 ± 5.88
HD11	1.058 ± 0.165	147.45 ± 3.68
HD32	1.236 ± 0.166	260.03 ± 19.29
HD41	1.538 ± 0.138	539.00 ± 36.86

4. Conclusions

We reported on the fabrication of pH-responsive hydrogels with high conversion degree from HEMA and DEAEMA monomer precursors directly on Si substrates via non-thermal atmospheric plasma polymerization. The process is simple, only bulk monomers in nanoliter volume and a plasma jet are needed, and it allows fast fabrication (30 s) of the hydrogel coatings, thus, avoiding tedious multi-step preparation. As no organic waste is produced during the synthesis, the mechanism is environmental friendly. The chemical and physical properties of the hydrogels can be tuned by simply varying the mixture ratio. DEAEMA is pH responsive due to the presence of the tertiary amine which can be protonated and deprotonated in acidic and alkaline, respectively. Therefore, the pH responsive behavior of the hydrogels can be modulated by increasing the DEAMEA content in the mixture. The wrinkles characteristics, such as wrinkle wavelength and wrinkle amplitude/ height, depend on the total thickness of the hydrogel, which is also strongly influenced by the mixture ratio. The wrinkle wavelength and amplitude are increasing with increasing content of HEMA. These chemical and physical properties of the hydrogels are preserved even after immersion in water for several days indicating the high stability. The ability to prepare unique materials with tunable properties will certainly open a plethora of applications.

CRediT authorship contribution statement

Monique Levien: Conceptualization, Methodology, Writing – original draft, Writing – review & editing. **Ihsan Amin:** Conceptualization, Methodology, Writing – original draft, Writing – review & editing. **Felipe Vicente de Paula Kodaira:** Methodology, Writing – original draft, Writing – review & editing. **Klaus-Dieter Weltmann:** Methodology, Writing – original draft, Writing – review & editing, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

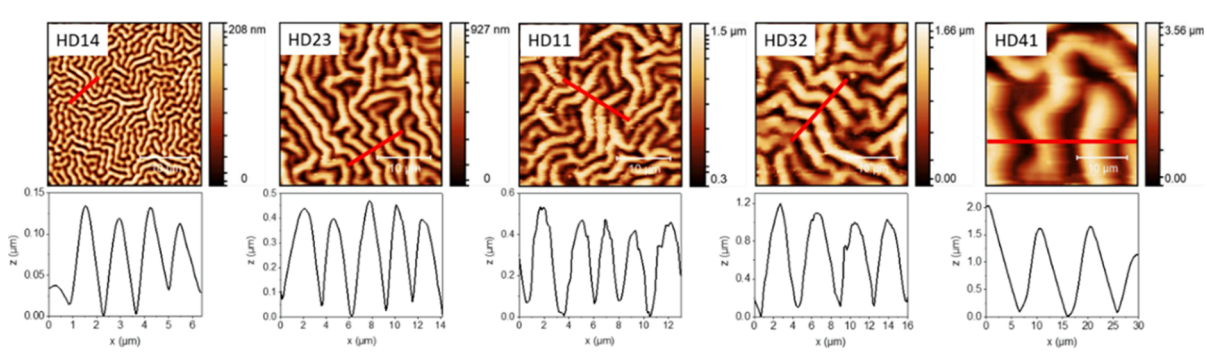

Fig. 6. AFM images of as-deposited HD films depending on the HD mixture ratio and corresponding height profiles.

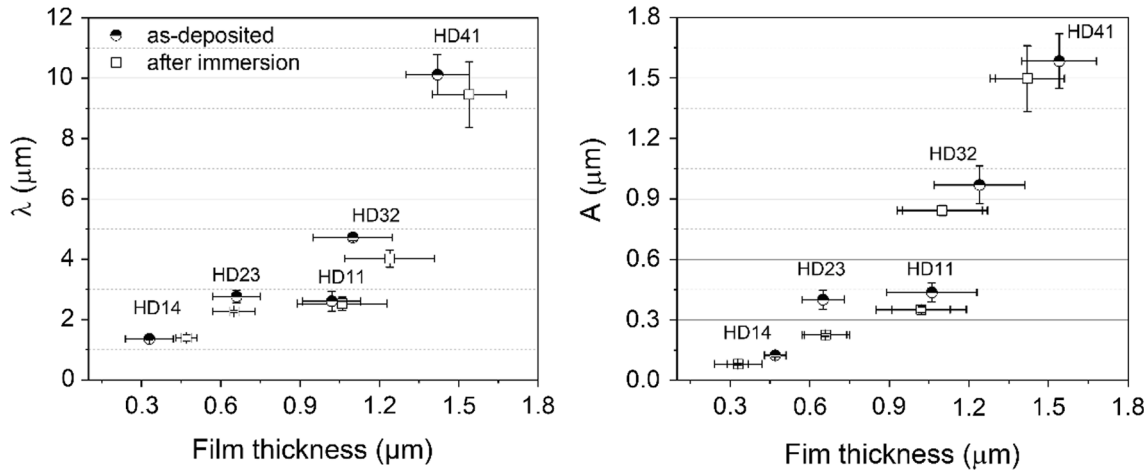

Fig. 7. Wrinkle wavelength (λ) and wrinkle height (A) depending on the film thickness determined by AFM analysis.

Data availability

Data will be made available on request.

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The raw/processed data required to reproduce these findings can be shared on demand.

Appendix A. Supplementary material

Supplementary data to this article can be found online at https://doi.org/10.1016/j.eurpolymj.2023.112413.

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